

Section - I: Multiple Choice Questions

1. What is a common indication of overfitting in a machine learning model?
 - A) The model performs well on both the training and testing datasets.
 - B) The model performs well on the training dataset but poorly on the testing dataset.
 - C) The model performs poorly on both the training and testing datasets.
 - D) The model is unable to learn from the training dataset.

Section - II: Descriptive / Case Study-Based Questions

1. A marketing company, GreenTech, wants to develop a machine learning model to predict customer churn based on their usage patterns and demographic data. The model will be trained on a dataset of 10,000 customers, with 8,000 used for training and 2,000 for testing. After training several models, the data scientist observes the following results: Model 1: Training accuracy = 98%, Testing accuracy = 60% Model 2: Training accuracy = 80%, Testing accuracy = 75% Which type of problem is Model 1 experiencing, and which type of problem is Model 2 experiencing? Explain your answers and suggest possible solutions to improve the performance of each model.